MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, 262,144 BIT, MOS, ULTRAVIOLET ERASABLE PROGRAMMABLE READ-ONLY MEMORY (EPROM) MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon N-channel MOS erasable programmable read-only memory microcircuits which employ the ultraviolet light erasing technique. Two product assurance classes and a choice of case outline and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	32,768 8 bit EPROM (250 ns)
02	32,768 8 bit EPROM (200 ns)
03	32,768 8 bit EPROM (170 ns)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
Y	D-10 (28-lead, 1/2" x 1-3/8"), dual-in-line package Transparent lid to permit UV light erasure
Z	C-12 (32-terminal RECT. CC. 450" x .550") Transparent lid to permit UV light erasure

1.3 Absolute maximum ratings.

Supply voltage range (V_{CC}) 1/	-0.6 V dc to 6.25 V dc
All input or output voltages	-0.6 V dc to 6.25 V dc
Program input (Vpp)	-0.6 V dc to 13.0 V dc
Operating case temperature range (T_C)	-55°C to +125°C
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	+260°C
Thermal resistance, junction-to-case (θ_{JC})	30°C/W
Maximum power dissipation (P_D)	1.0 watt
Junction temperature (T_J) 2/	+175°C

1/ All voltages referenced to V_{SS}.

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 C dc
High-level program input voltage (V_{PP})	12.5 V
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.2.1 Unprogrammed or erased devices. The truth table for unprogrammed devices shall be as specified on figure 2.

3.2.2.2 Programmed devices. The requirements for supplying programmed devices are not part of this specificaton.

3.2.3 Functional block diagram. The functional block diagram shall be as specified on figure 3. Upon implementing design changes, the manufacturer must submit a new block diagram to the qualifying activity for inclusion in this specification. The block diagram shall clearly define the row address inputs and the column address inputs.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

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TABLE I. Electrical performance characteristics (device types 01, 02, 03).

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level input voltage	V_{IH}		All	2.0	V_{CC} +1	V
Low-level input voltage	V_{IL}		All	-0.1	0.8	V
High-level output voltage	V_{OH}	V_{CC} = 4.5 V I_{OH} = -400 µA	All	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V I_{OL} = 2.1 mA	All		0.45	V
Output leakage current	I_{OHZ}	V_O = 2.4 V; V_{CC} = 5.5 V	All	-10	+10	µA
	I_{OLZ}	V_O = 0.4 V				
High-level input current	I_{IH}	Address, $\overline{CE}$, $\overline{OE}$ V_{CC} = 4.5 V to 5.5 V V_{IN} = V_{CC}	All	-10	+10	µA
Low-level input current	I_{IL}	Address, $\overline{CE}$, $\overline{OE}$ V_{CC} = 5.5 V V_{IN} = 0.0 V	All	-10	+10	µA
Supply current (standby)	I_{CC1}	$\overline{CE}$ = V_{IH} $\overline{OE}$ = V_{IL} V_{CC} = 5.5 V Output open	All		50	mA
Supply current (active)	I_{CC2}	$\overline{CE}$ = V_{IL} V_{CC} = 5.5 V Output open	01,02		125	mA
			03		140	mA
V_{pp} current read/standby	I_{PP1}	V_{pp} = 5.5 V	All		5	mA
Input capacitance	C_i	V_I = 0 V, f = 1 MHz	All		6	pF
Output capacitance	C_o	V_O = 0 V, f = 1 MHz	All		12	pF
Margin voltage	V_m	HMOS IIE	All	5.9		V
Address inputs to outputs (access time)	t_{AVQV}	$\overline{CE}$ = $\overline{OE}$ = V_{IL} V_{CC} = 4.5 V to 5.5 V (see figure 4)	01		250	ns
			02		200	ns
			03		170	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics (device types 01, 02, 03) - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Chip enable to outputs (access time)	t_{ELQV}	$\overline{OE} = V_{IL}$ V_{CC} = 4.5 V to 5.5 V (see figure 4)	01		250	ns
			02		200	ns
			03		170	ns
Output enable to outputs (access time)	t_{OLQV}	$\overline{CE} = V_{IL}$ V_{CC} = 4.5 V to 5.5 V (see figure 4)	01		130	ns
			02,03		75	ns
Address to outputs hold	t_{AXQX}	$\overline{CE} = \overline{OE} = V_{IL}$ V_{CC} = 4.5 V to 5.5 V (see figure 4)	All	0		ns
Output enable high to output float	t_{OHQZ}	V_{CC} = 4.5 V to 5.5 V (see figure 4)	All	0	55	ns
Chip enable high to output float	t_{EHQZ}	V_{CC} = 4.5 V to 5.5 V (see figure 4)	All	0	100	ns

1/ Complete terminal conditions shall be specified in table III.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) 1/ 2/ 3/ 4/ 5/	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	2,8,10	2,8,10
Final electrical test parameters (method 5004)	1*,2,3,7*,9,10,11	1*,2,3,7*,8,9,10,11
Group A test requirements (method 5005)	1,2,3,4,7,8,9,10,11	1,2,3,4,7,8,9,10,11
Group B test requirements (method 5005)	1,2,3,4,7,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	2,8,10
Group D end-point electrical parameters (method 5005)	2,8,10	2,8,10

1/ (*)PDA applies to subgroup 1 and 7 (see 4.2c).
2/ Any or all subgroups may be combined when using high-speed testers.
3/ Subgroup 7 shall consist of verifying the specified pattern.
4/ For all electrical tests, the device shall be programmed to the pattern specified.
5/ Test number 8 in subgroup S and B are to be run hot only.

3.7 Processing EPROMS. All testing requirements and quality assurance provisions herein shall be satisfied by the manufacturer prior to delivery.

3.7.1 Erasure of EPROMS. When specified, devices shall be erased in accordance with the procedure and characteristics specified in 4.8.

3.7.2 Programmability of EPROMS. When specified, device shall be programmed to the specified pattern using the procedures and characteristics specified in 4.7 and table IV.

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3.7.3 Verification of erasure or programmability of EPROMS. When specified, devices shall be verified as either programmed to the specified pattern or erased. As a minimum, verification shall consist of performing a functional test (subgroup 7) to verify that all bits are in the proper state. Any bit that does not verify to be in the proper state shall constitute a device failure, and shall be removed from the lot.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 47 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical tests shall be as specified in table II, except interim electrical parameters tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 5 percent based on failures from group A, subgroups 1 and 7, test after cooldown as final electrial parameter tests are performed prior to burn-in, failures resulting from pre-burn-in screening may be excluded from the PDA. The verified failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. EPROM data retention test flow.

(1) Program (i.e., charge the floating gate) a minimum of 95 percent of the bit locations, including the slowest programming cell (manufacturer to provide information on which are the slowest programming cells for a particular design). The remaining cells shall provide a worst case speed pattern.

(2) Bake, unbiased, for 72 hours at +140°C to screen for data retention lifetime.

(3) Perform "raw class" testing (+25°C), which includes a margin test. For the margin test, use V_m = +5.9 V and loose timing ($t_{ACC} \geq 1$ µs).

(4) Perform a dynamic burn-in per MIL-STD-883.

(5) Perform post-burn in electrical checks at +25°C and +125°C, to include margin testing (V_m = +5.9 V, $t_{ACC} \geq 1$ µs).

(6) Erase device; then program the remaining bits which were not programmed in step 1 above.

(7) Perform electrical testing at -55°C, to include a margin test (V_m = +5.9 V, $t_{ACC} \geq$ 1 µs).

(8) Erase device; perform erase check.

(9) Mark and package.

(10) Perform final quality assurance procedures.

e. After completion of all testing, the devices shall be erased and verified prior to delivery (except devices submitted for groups A, B, C, and D testing).

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Qualification data for subgroups 7, 9, 10, and 11 shall be attributes only.

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies one device type which is manufactured identically to other device type on this specificaton, then the other device types may be part I qualified by conducting only group A electrical tests and submitting data in accordance with MIL-M-38510, appendix D (i.e. groups B, C, and D tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Half the devices selected for testing shall be programmed with the pattern shown on figure 7. The remaining devices shall be programmed with the complement of the pattern shown on figure 7. After completion of all testing, the devices shall be erased and verified (except submitted for group B, C, and D testing).

d. Subgroup 4 (C_i and C_o measurements) shall be measured only for initial qualification and after process or design change which may affect capacitance.

e. Subgroup 12 shall be added to group A inspection requirements for class B devices using an LTPD of 10 and consist of procedures, test conditions and limits specified in table III.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

a. A special subgroup shall be added using an LTPD of 15 for classes S and B. This subgroup shall consist of a high voltage test of the input protection circuits, and shall be tested per MIL-STD-883C, method 3015.

b. Steady-state life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using the circuit on figure 5.

c. Half the devices selected for class S electrical testing shall be programmed with the pattern shown on figure 7. The remaining devices shall be programmed with the complement of the pattern shown on figure 7. After completion of all testing, the devices shall be erased and verified (except devices submitted for group C and D testing).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

(4) The devices selected for testing shall be programmed with a random pattern (i.e., data programmed should exhibit no pattern dependency). After completion of all testing, the devices shall be erased and verified (except devices submitted for group D testing).

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c. A reprogrammability test shall be added to group C inspection prior to performing the steady-state life test (see 4.4.3b) using an LTPD of 10. The devices to be submitted to the steady-state life testing shall be subjected to the following tests and examinations:

(1) Each device in the sample shall be subjects to a minimum of 50 program and erase cycles. Each cycle shall consist of the following steps:

a. Program half the devices with the pattern shown on figure 7 program the remaining devices with the complement of the pattern shown on figure 7.

b. Verify patterns (see 3.7.3).

c. Erase (see 3.7.1).

d. Verify pattern erasure (see 3.7.3).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883.

a. End-point electrical parameters shall be as specified in table II herein.

b. The devices selected for testing shall be programmed with a random patern (i.e., data programmed should exhibit no pattern dependency). After completion of all testing, the devices shall be erased and verified (except devices submitted for group D testing).

4.4.5 Constant acceleration. The constant acceleration tests of groups C and D shall be performed using test condition D.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-M-38510.

4.7 Programming procedure. The following procedure shall be followed when programming and verification testing is performed. The waveforms and timing relationships shown is figure 6 and test conditions and limits specified in table IV shall be adhered to:

a. Initially, and after each erasure, all bits are in the "H" state (output high). Information is introduced by selectively programming "L" into the desired bit locations. A programmed "L" can be changed to an "H" by ultraviolet light erasure.

b. The programming algorithm (see figure 11) utilizes two different pulse types: Initial and overprogram. The duration of the initial CE pulse(s) is one millisecond which will then be followed by a longer overprogram pulse of length 3X millisecond. Where X is an iteration counter and is equal to the number of the initial one-millisecond pulses applied to a particular addressed location, before a correct verify occurs. Up to 25 one-millisecond pulses per byte are provided for, before the overprogram pulse is applied.

c. A verify should be performed on the programmed bits to determine that they were correctly programmed. The entire sequence of program pulses and byte verifications is performed at V_{CC} = 6.0 V and V_{pp} = 12.5 V. When the programming cycle has been completed, all bytes should be compared to the original data with V_{CC} = V_{pp} = 5.0 V..

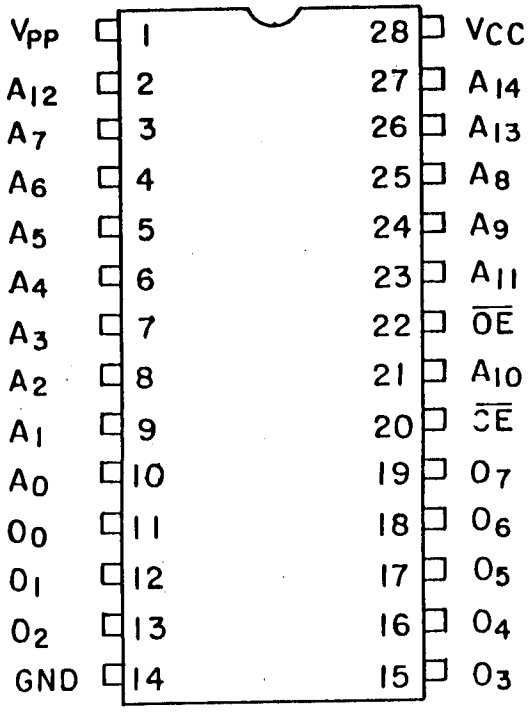
4.8 Erasing procedure. The device is erased by exposure to high intensity short wave ultraviolet light at a wavelength to 253.7 nm. The recommended integrated dose (i.e., UV intensity X exposure time) is a minimum of 15 W-s/cm^2. The EPROM should be placed about 1 inch away from the light source for a period of 20 minutes. After erasure all bits are in the high state.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. The devices covered by this specification require electrostatic protection.

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Device types 01, 02, and 03

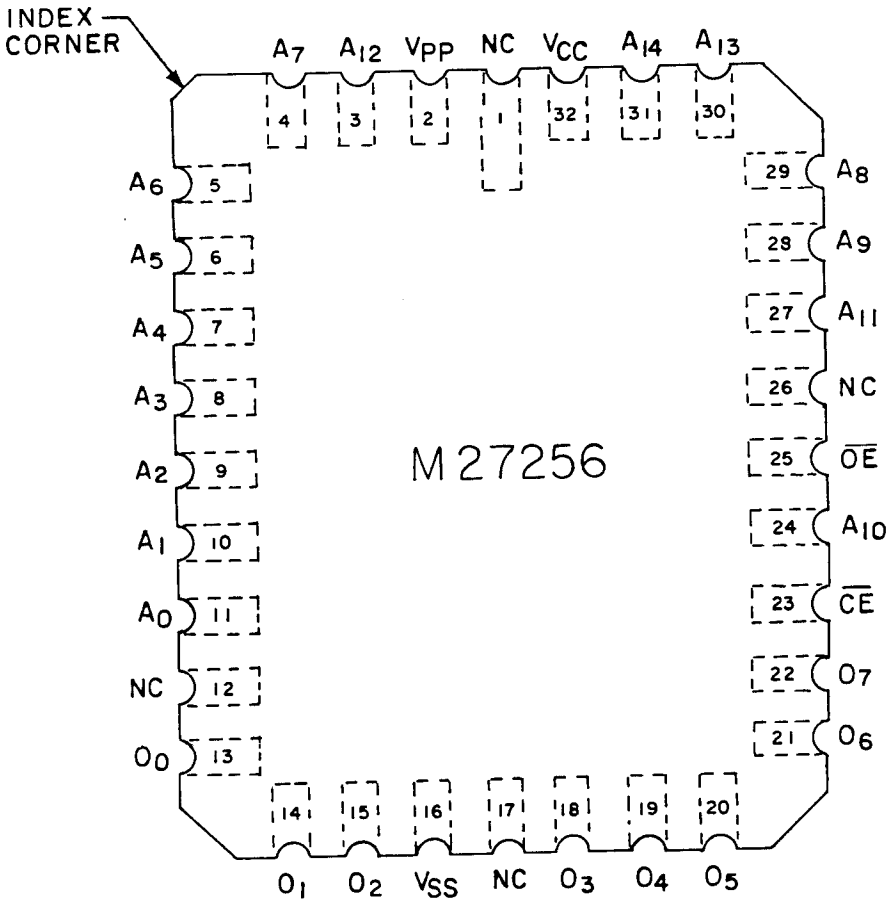


Case Y

FIGURE 1. Terminal connections.

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Device types 01, 02, and 03



Case Z

FIGURE 1. Terminal connections - Continued.

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Unprogrammed

Word in	Inputs				Outputs
	$\overline{CE}$	$\overline{OE}$	V_{PP}	A14-A0	O0-O7
X	V_{IL}	V_{IL}	V_{CC}	X	V_{OH}
X	V_{IH}	X	V_{CC}	X	HI-Z
X	X	V_{IH}	V_{CC}	X	HI-Z

X = Irrelevant
HI-Z = High-impedance state

Mode selection

Mode	Inputs				Outputs
	$\overline{CE}$	$\overline{OE}$	A9	V_{PP}	O0-O7
Read	V_{IL}	V_{IL}	X	V_{CC}	D_{out}
Output disable	V_{IL}	V_{IH}	X	V_{CC}	HI-Z
Standby	V_{IH}	X	X	V_{CC}	HI-Z
Program	V_{IL}	V_{IH}	X	V_{PP}	D_{in}
Verify	V_{IH}	V_{IL}	X	V_{PP}	D_{out}
Optional verify	V_{IL}	V_{IL}	X	V_{PP}	D_{out}
Program inhibit	V_{IH}	V_{IH}	X	V_{PP}	HI-Z
Indentifier	V_{IL}	V_{IL}	V_H	V_{CC}	Code

X = Irrelevant
HI-Z = High-impedance state
V_H = 12.0 V +/-0.5 V

FIGURE 2. Truth tables.

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Device types 01, 02, and 03

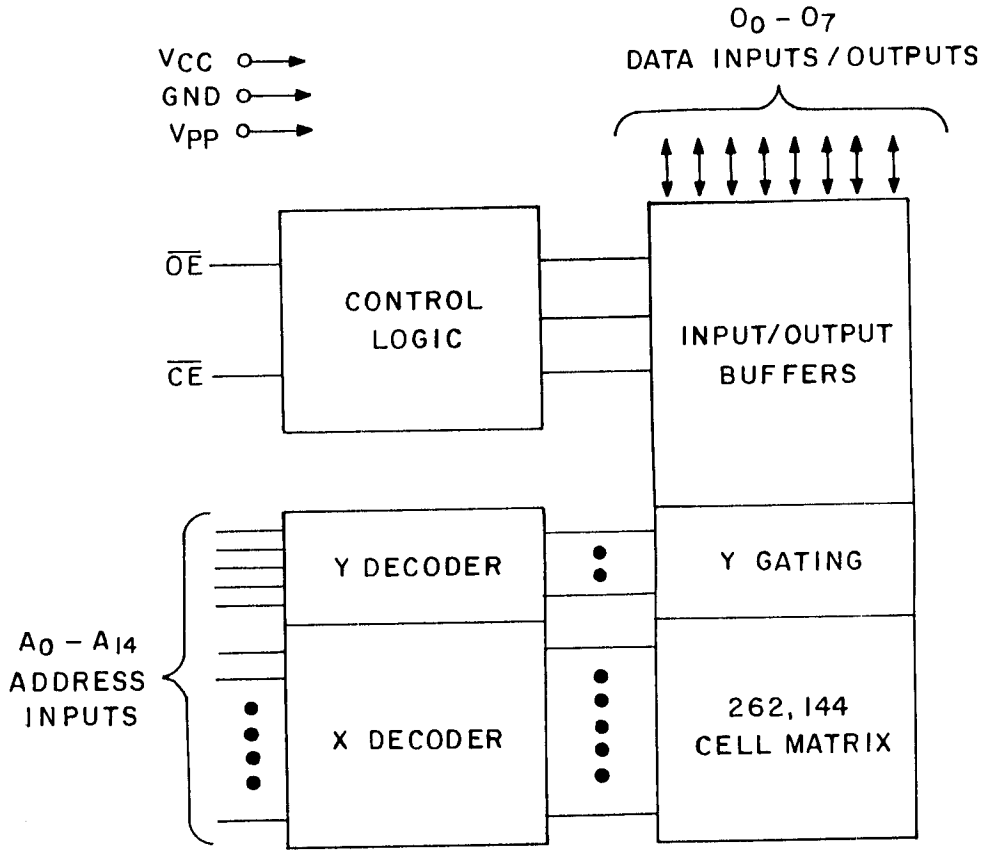
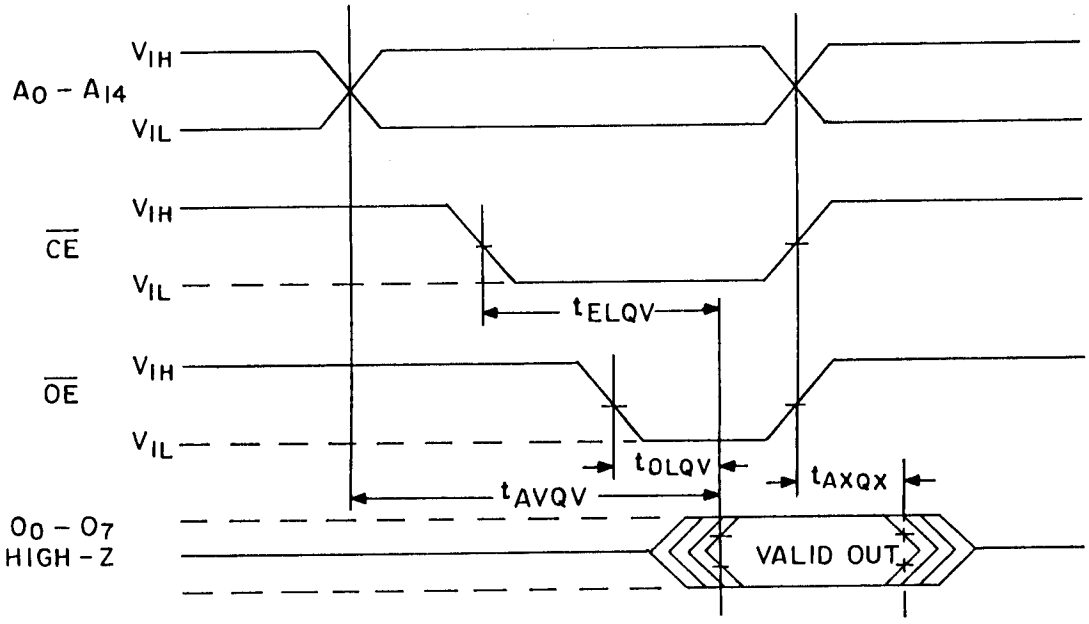


FIGURE 3. Functional block diagram.

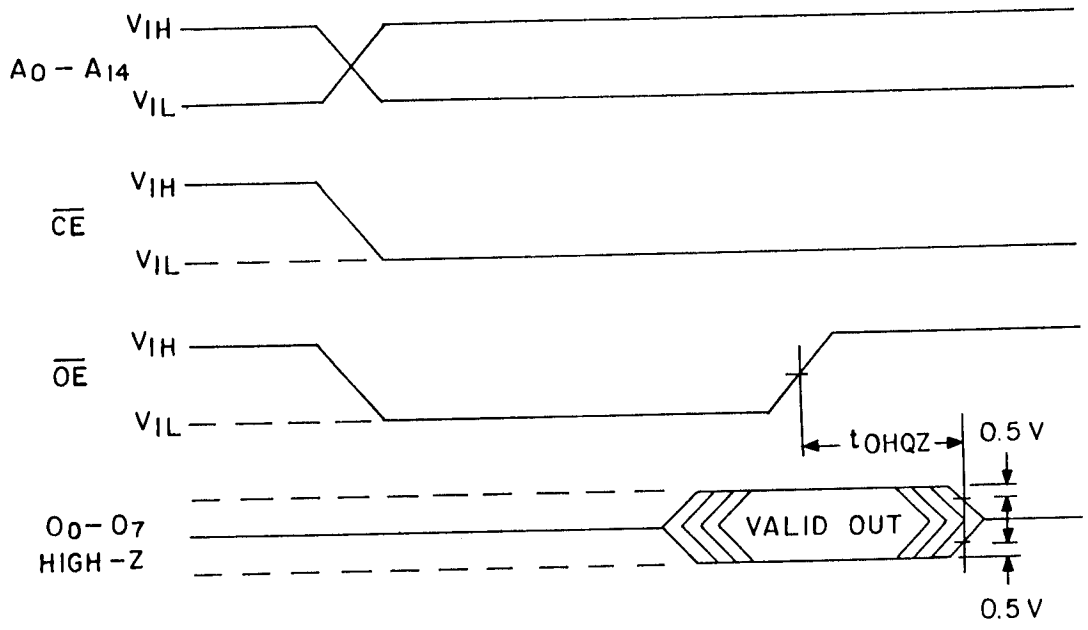
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Read timing

FIGURE 4. A.C. Waveforms.

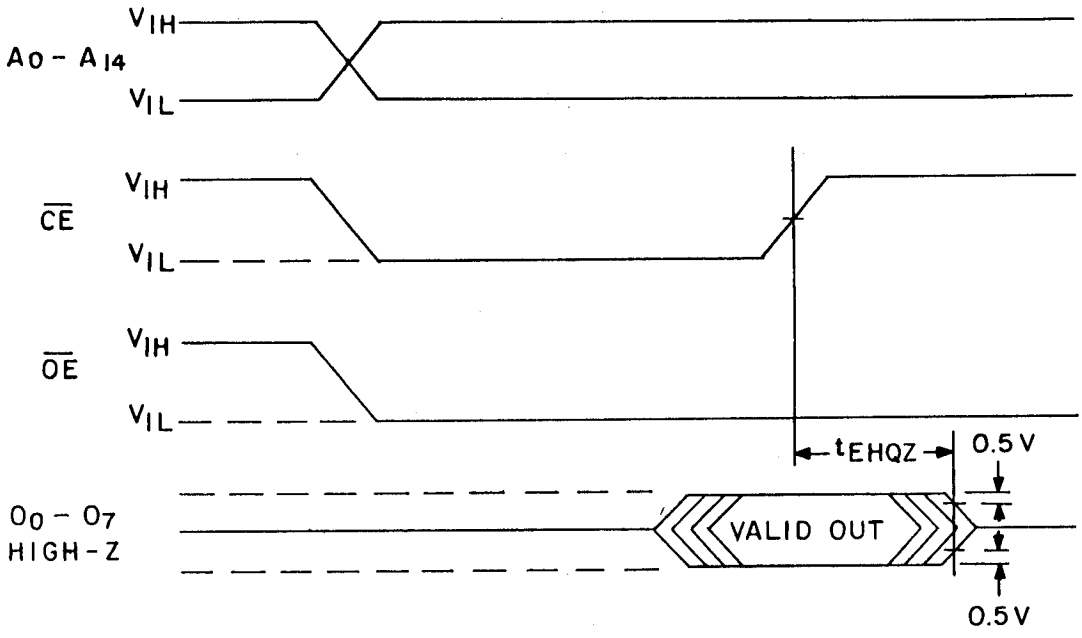
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Output disable timing

FIGURE 4. A.C. Waveforms -Continued.

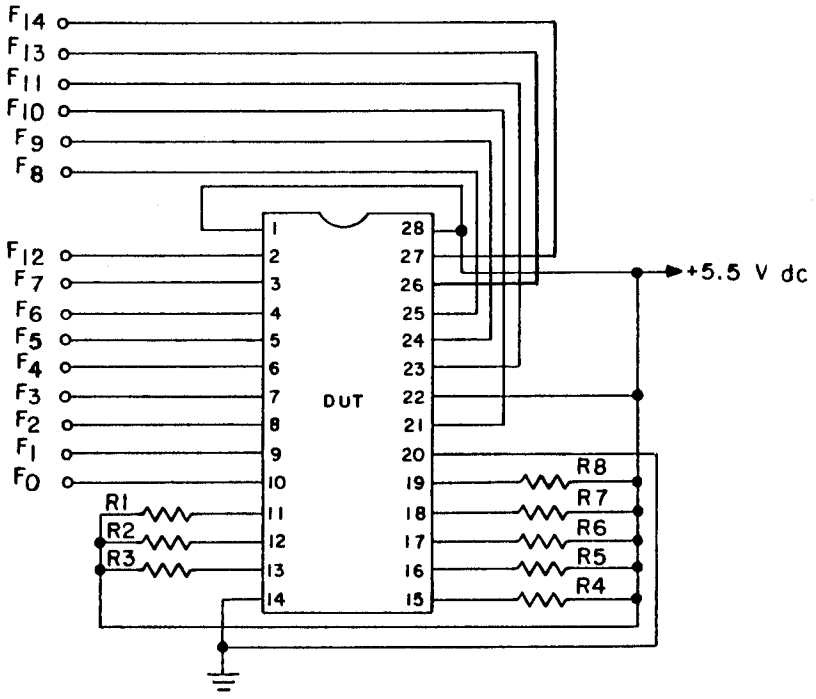
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Output disable timing

FIGURE 4. A.C. Waveforms -Continued.

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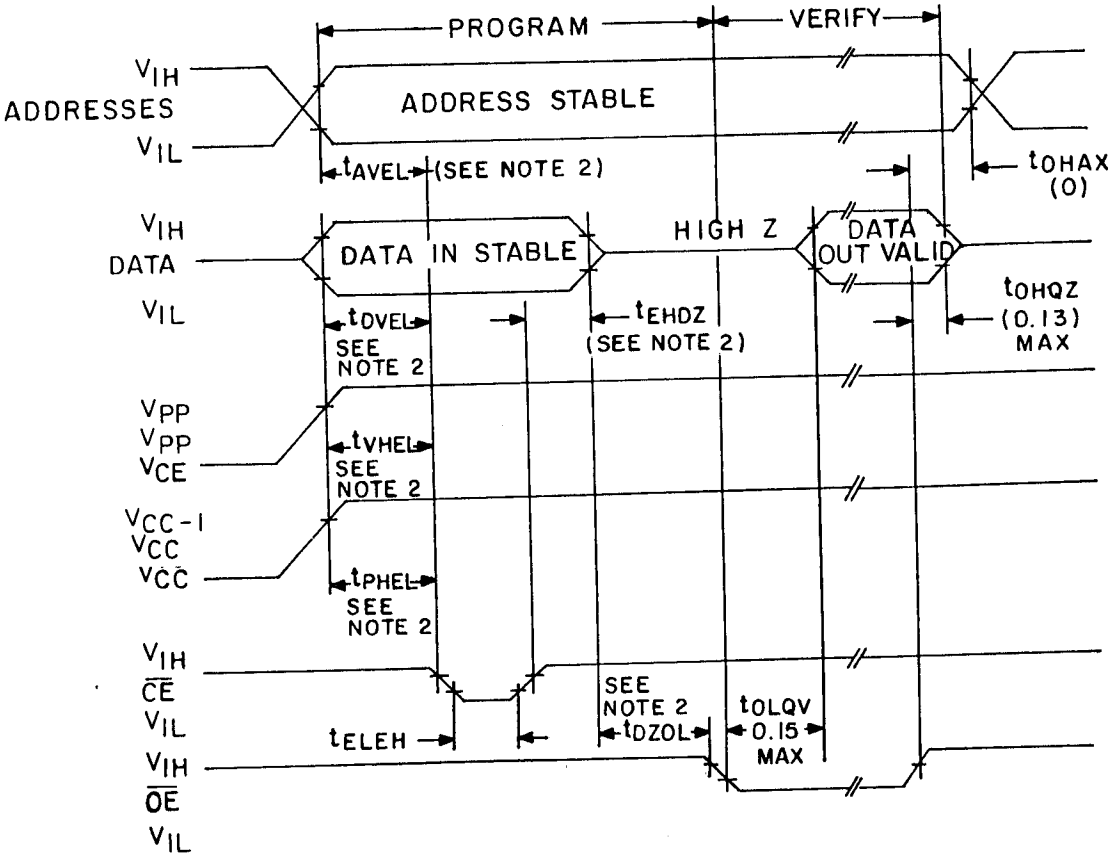


NOTES:
1. R1 thru R8 = 1.0 k Ω +/-5.0%.
2. All pulse generators have the following characteristics:
   V_{IL} = 0.0 V minimum to 0.65 V maximum
   V_{IH} = 2.2 V minimum to 5.50 V maximum
   50%+ /-15% duty cycle and frequencies as specified in note 3.
3. Input frequencies are as follows:
   frequency (+/- 20%)

F_0 = 16 kHz < F_0 < 100 kHz
F_1 = 1/2 F_0
F_2 = 1/2 F_1
F_3 = 1/2 F_2
F_4 = 1/2 F_3
F_5 = 1/2 F_4
F_6 = 1/2 F_5
F_7 = 1/2 F_6
F_8 = 1/2 F_7
F_9 = 1/2 F_8
F_{10} = 1/2 F_9
F_{11} = 1/2 F_{10}
F_{12} = 1/2 F_{11}
F_{13} = 1/2 F_{12}
F_{14} = 1/2 F_{13}

FIGURE 5. Burn-in and steady state life test circuit.

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NOTES:
1. All time shown in ( ) are minimum and in µsec unless otherwise specified.
2. The input timing reference level is 0.8 V for a V_{IL} and 2 V for a V_{IH}.
3. t_{OLQV} and t_{HQZ} are characteristics of the device but must be accommodated by the programmer.
4. When programming the device a 0.1 µF capacitior is required across Vpp and GND to suppress spurious voltage transients which can damage the device.

FIGURE 6. Programming waveforms.

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TABLE III. Group A inspection. 1/ 2/

Subgroup	Symbol	MIL-STD-883-method	Case Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14
			Case Z	2	3	4	5	6	7	8	9	10	11	13	14	15	16
			Test no.	V_{PP}	A12	A7	A6	A5	A4	A3	A2	A1	A0	O0	O1	O2	GND
1 T_C = +25°C	V_{OH}	3006	1	4.5 V	3/	3/	3/	3/	3/	3/	3/	3/	3/	1.4 mA			
			2	"	"	"	"	"	"	"	"	"	"	"	.4 mA		
			3	"	"	"	"	"	"	"	"	"	"	"		.4 mA	
			4	"	"	"	"	"	"	"	"	"	"	"			1.4 mA
			5	"	"	"	"	"	"	"	"	"	"	"			
			6	"	"	"	"	"	"	"	"	"	"	"			
			7	"	"	"	"	"	"	"	"	"	"	"			
			8	"	"	"	"	"	"	"	"	"	"	"			
	V_{OL}	3007	9	"	"	"	"	"	"	"	"	"	"	2.1 mA			
			10	"	"	"	"	"	"	"	"	"	"	"	2.1 mA		
			11	"	"	"	"	"	"	"	"	"	"	"		2.1 mA	
			12	"	"	"	"	"	"	"	"	"	"	"			
			13	"	"	"	"	"	"	"	"	"	"	"			
			14	"	"	"	"	"	"	"	"	"	"	"			
			15	"	"	"	"	"	"	"	"	"	"	"			
			16	"	"	"	"	"	"	"	"	"	"	"			
	I_{OHZ}		17	5.5 V	4/	4/	4/	4/	4/	4/	4/	4/	4/	2.4 V			
			18	"	"	"	"	"	"	"	"	"	"	"	2.4 V		
			19	"	"	"	"	"	"	"	"	"	"	"		2.4 V	
			20	"	"	"	"	"	"	"	"	"	"	"			2.4 V
			21	"	"	"	"	"	"	"	"	"	"	"			
			22	"	"	"	"	"	"	"	"	"	"	"			
			23	"	"	"	"	"	"	"	"	"	"	"			
			24	"	"	"	"	"	"	"	"	"	"	"			
	I_{OLZ}		25	"	"	"	"	"	"	"	"	"	"	.45 V			
			26	"	"	"	"	"	"	"	"	"	"	"	.45 V		
			27	"	"	"	"	"	"	"	"	"	"	"		.45 V	
			28	"	"	"	"	"	"	"	"	"	"	"			.45 V
			29	"	"	"	"	"	"	"	"	"	"	"			
			30	"	"	"	"	"	"	"	"	"	"	"			
			31	"	"	"	"	"	"	"	"	"	"	"			
			32	"	"	"	"	"	"	"	"	"	"	"			
			33	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"			
			34	"	"	"	"	"	"	"	"	"	"	"			
			35	"	"	"	"	"	"	"	"	"	"	"			
			36	"	"	"	"	"	"	"	"	"	"	"			
			37	"	"	"	"	"	"	"	"	"	"	"			
			38	"	"	"	"	"	"	"	"	"	"	"			
			39	"	"	"	"	"	"	"	"	"	"	"			
			40	"	"	"	"	"	"	"	"	"	"	"			
			41	"	"	"	"	"	"	"	"	"	"	"			
			42	"	"	"	"	"	"	"	"	"	"	"			
			43	"	"	"	"	"	"	"	"	"	"	"			
			44	"	"	"	"	"	"	"	"	"	"	"			
			45	"	"	"	"	"	"	"	"	"	"	"			
			46	"	"	"	"	"	"	"	"	"	"	"			
			47	"	"	"	"	"	"	"	"	"	"	"			
			48	"	"	"	"	"	"	"	"	"	"	"			
			49	"	"	"	"	"	"	"	"	"	"	"			
	I_{IL}	3009	50	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"			
			51	"	"	"	"	"	"	"	"	"	"	"			
			52	"	"	"	"	"	"	"	"	"	"	"			
			53	"	"	"	"	"	"	"	"	"	"	"			
			54	"	"	"	"	"	"	"	"	"	"	"			
			55	"	"	"	"	"	"	"	"	"	"	"			
			56	"	"	"	"	"	"	"	"	"	"	"			
			57	"	"	"	"	"	"	"	"	"	"	"			
			58	"	"	"	"	"	"	"	"	"	"	"			
			59	"	"	"	"	"	"	"	"	"	"	"			
			60	"	"	"	"	"	"	"	"	"	"	"			
			61	"	"	"	"	"	"	"	"	"	"	"			
			62	"	"	"	"	"	"	"	"	"	"	"			
			63	"	"	"	"	"	"	"	"	"	"	"			
			64	"	"	"	"	"	"	"	"	"	"	"			
			65	"	"	"	"	"	"	"	"	"	"	"			
			66	"	"	"	"	"	"	"	"	"	"	"			

Subgroup	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Measured terminal	Test limits Min	Test limits Max	Unit
	18	19	20	21	22	23	24	25	27	28	29	30	31	32				
	O3	O4	O5	O6	O7	$\overline{CE}$	A10	$\overline{OE}$	A11	A9	A8	A13	A14	V_{CC}				
1 T_C = +25°C	.4 mA	.4 mA	.4 mA	.4 mA	.4 mA	0.8 V	3/	3/	3/	3/	3/	3/	3/	4.5 V	O0	2.4		V
						"	"	"	"	"	"	"	"	"	O1	"		"
						"	"	"	"	"	"	"	"	"	O2	"		"
						"	"	"	"	"	"	"	"	"	O3	"		"
						"	"	"	"	"	"	"	"	"	O4	"		"
						"	"	"	"	"	"	"	"	"	O5	"		"
						"	"	"	"	"	"	"	"	"	O6	"		"
						"	"	"	"	"	"	"	"	"	O7	"		"
	2.1 mA	2.1 mA	2.1 mA	2.1 mA	2.1 mA	"	"	"	"	"	"	"	"	"	O0		0.45	"
						"	"	"	"	"	"	"	"	"	O1		"	"
						"	"	"	"	"	"	"	"	"	O2		"	"
						"	"	"	"	"	"	"	"	"	O3		"	"
						"	"	"	"	"	"	"	"	"	O4		"	"
						"	"	"	"	"	"	"	"	"	O5		"	"
						"	"	"	"	"	"	"	"	"	O6		"	"
						"	"	"	"	"	"	"	"	"	O7		"	"
	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	4/	4/	4/	4/	4/	4/	4/	4/	5.5 V	O0	-10	+10	µA
						"	"	"	"	"	"	"	"	"	O1	"	"	"
						"	"	"	"	"	"	"	"	"	O2	"	"	"
						"	"	"	"	"	"	"	"	"	O3	"	"	"
						"	"	"	"	"	"	"	"	"	O4	"	"	"
						"	"	"	"	"	"	"	"	"	O5	"	"	"
						"	"	"	"	"	"	"	"	"	O6	"	"	"
						"	"	"	"	"	"	"	"	"	O7	"	"	"
	.45 V	.45 V	.45 V	.45 V	.45 V	"	"	"	"	"	"	"	"	"	O0	"	"	"
						"	"	"	"	"	"	"	"	"	O1	"	"	"
						"	"	"	"	"	"	"	"	"	O2	"	"	"
						"	"	"	"	"	"	"	"	"	O3	"	"	"
						"	"	"	"	"	"	"	"	"	O4	"	"	"
						"	"	"	"	"	"	"	"	"	O5	"	"	"
						"	"	"	"	"	"	"	"	"	O6	"	"	"
						"	"	"	"	"	"	"	"	"	O7	"	"	"
						"	"	"	"	"	"	"	"	"	A12	"	"	"
						"	"	"	"	"	"	"	"	"	A7	"	"	"
						"	"	"	"	"	"	"	"	"	A6	"	"	"
						"	"	"	"	"	"	"	"	"	A5	"	"	"
						"	"	"	"	"	"	"	"	"	A4	"	"	"
						"	"	"	"	"	"	"	"	"	A3	"	"	"
						"	"	"	"	"	"	"	"	"	A2	"	"	"
						"	"	"	"	"	"	"	"	"	A1	"	"	"
						"	"	"	"	"	"	"	"	"	A0	"	"	"
						5.5 V	"	"	"	"	"	"	"	"	$\overline{CE}$	"	"	"
						"	5.5 V	"	"	"	"	"	"	"	A10	"	"	"
						"	"	5.5 V	"	"	"	"	"	"	$\overline{OE}$	"	"	"
						"	"	"	5.5 V	"	"	"	"	"	A11	"	"	"
						"	"	"	"	5.5 V	"	"	"	"	A9	"	"	"
						"	"	"	"	"	5.5 V	"	"	"	A8	"	"	"
						"	"	"	"	"	"	5.5 V	"	"	A13	"	"	"
						"	"	"	"	"	"	"	5.5 V	"	A14	"	"	"
						"	"	"	"	"	"	"	"	"	A12	"	"	"
						"	"	"	"	"	"	"	"	"	A7	"	"	"
						"	"	"	"	"	"	"	"	"	A6	"	"	"
						"	"	"	"	"	"	"	"	"	A5	"	"	"
						"	"	"	"	"	"	"	"	"	A4	"	"	"
						"	"	"	"	"	"	"	"	"	A3	"	"	"
						"	"	"	"	"	"	"	"	"	A2	"	"	"
						"	"	"	"	"	"	"	"	"	A1	"	"	"
						"	"	"	"	"	"	"	"	"	A0	"	"	"
						0.0 V	"	"	"	"	"	"	"	"	$\overline{CE}$	"	"	"
						"	0.0 V	"	"	"	"	"	"	"	A10	"	"	"
						"	"	0.0 V	"	"	"	"	"	"	$\overline{OE}$	"	"	"
						"	"	"	0.0 V	"	"	"	"	"	A11	"	"	"
						"	"	"	"	0.0 V	"	"	"	"	A9	"	"	"
						"	"	"	"	"	0.0 V	"	"	"	A8	"	"	"
						"	"	"	"	"	"	0.0 V	"	"	A13	"	"	"
						"	"	"	"	"	"	"	0.0 V	"	A14	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Mil-STD-883-method	Case Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Measured terminal	Test limits		Unit
			Case Z	2	3	4	5	6	7	8	9	10	11	13	14	15	16	18	19	20	21	22	23	24	25	27	28	29	30	31	32		Min	Max	
			Test no.	V_{PP}	A12	A7	A6	A5	A4	A3	A2	A1	A0	O0	O1	O2	GND	O3	O4	O5	O6	O7	$\overline{CE}$	A10	$\overline{OE}$	A11	A9	A8	A13	A14	V_{CC}				
1 T_C = +25°C	I_{CC1}	3005	67	5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND				GND						2.0 V	GND	2.0 V	GND	GND	GND	GND	GND	5.5 V	V_{CC}		50	mA
	I_{CC2} 01 02 03		68	"	GND	GND	GND	GND	GND	GND	GND	GND	GND				"						.45 V	GND	.45 V	GND	GND	GND	GND	GND	"	V_{CC}		125 125 140	mA " " "
	I_{PP1}		69	"	GND	GND	GND	GND	GND	GND	GND	GND	GND				"						2.0 V	GND	.45 V	GND	GND	GND	GND	GND	"	V_{PP}		5	"
2, 3	Same tests, terminal conditions and limits as subgroup 1, except T_C = +125°C and -55°C.																																		
4	C_I	3012	70	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	All inputs		6	pF
	C_O		71	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	All outputs		12	pF
7	Functional tests	3014	72	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	.45 V	6/	.45 V	6/	6/	6/	6/	6/	6/	O0 thru O7		6/	
8	Same tests, terminal conditions and limits as subgroup 4, except T_C = +125°C.																																		
9	t_{AVQV} 01 02 03 t_{ELQV} 01 02 03 t_{GLQV} 01 02 03 t_{AXQX}		73 74 75 76	V_{CC} " " "	7/	7/	7/	7/	7/	7/	7/	7/	7/	8/	8/	8/	GND " " "	8/	8/	8/	8/	8/	7/	7/	7/	7/	7/	7/	7/	7/	7/	O0 thru O7 O	0	250 200 170 250 200 170 130 75 75	ns " " ns " " ns " "
10, 11	Same tests, terminal conditions and limits as subgroup 6, except T_C = +125°C and -55°C.																																		
12	t_{OHQZ}		77	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	GND	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	O0 thru O7		55	ns "

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1/ Terminal conditions:

a. Outputs not designated are open or resistively coupled to GND or voltage.

b. Inputs not designated are high > 2.0 V or low < 0.8 V.

2/ V_{CC} must be applied simultaneously or before V_{pp} and removed simutaneously or after V_{pp}.

3/ Before performing electrical tests, half the devices submitted for acceptance shall be programmed with the pattern shown on figure 7. The remaining devices shall be programmed with the complement of the pattern shown on figure 7. V_{IH} = 2.0 V and V_{IL} = 0.8 V shall be applied to the applicable address pins to set the measured terminal in the proper state for measurement.

4/ Terminal conditions for the output leakage current test shall be as follows:

a. V_{IH} = 2.0 V, V_{IL} = 0.8 V.

b. For I_{OLZ}, select an appropriate address to acquire a logic one on the designated output. Apply V_{IH} to $\overline{CE}$ and V_{IL} to $\overline{OE}$. Measure the leakage current while applying the specified voltage.

c. For I_{OHZ}, select an appropriate address to acquire a logic zero on the designated output. Apply V_{IL} to $\overline{CE}$ and V_{IH} to $\overline{OE}$. Measure the leakage current while applying the specified voltage.

5/ Input/output capacitance shall be measured between the designated terminal and the GND pin under the following conditions: $V_{CC} = V_I = V_O = 0V$, f = 1 MHz.

6/ Terminal conditions are as follows:

a. Inputs, see table V for V_{IL} and V_{IH} values.

b. Outputs, see table V for V_{OL} and V_{OH} values.

c. The functional tests shall be performed with V_{CC} = 4.5 V to V_{CC} = 5.5 V.

7/ For subgroups 9, 10 and 11, the address complement test shown on figures 8 and 9 shall be used. Input/output conditions appear in table V. Timing diagrams appear on figure 4. Subgroups 9, 10 and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

8/ The outputs are loaded in accordance with figure 10.

9/ For subgroup 12, input/output conditions appear in table V. Timing waveforms appear on figure 4. The outputs are loaded in accordance with figure 10. Additional terminal conditions are as follows:

a. For subgroup 12, V_{CC} = 4.5 V and V_{CC} = 5.5 V.

b. Output disable time from $\overline{CE}$ (see figure 4).

1. Output in logic low state - select an appropriate address to acquire a logic zero to the designated output $\overline{OE}$ = V_{IL}.

2. Output in logic high state - select an appropriate address to acquire a logic one to the designated output $\overline{OE}$ = V_{IL}.

c. Output disable time from $\overline{OE}$ (see figure 4).

1. Output in logic low state - select an appropriate address to acquire a logic zero to the designated output $\overline{CE}$ = V_{IL}.

2. Output in logic high state - select an appropriate address to acquire a logic one to the designated output $\overline{CE}$ = V_{IL}.

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TABLE IV. Programming characteristic.

Recommended timing requirements for programming, T_C = +25°C

Symbol	Parameter		Limits Min	Typ	Max	Unit
t_{ELEH}	Pulse width, program pulse $\overline{CE}$	Initial 1/	.95	1.0	1.05	ms
		Overprogram 2/	2.85		78.75	
t_{PHEL}	V_{CC} setup time		2			ms
t_{VHEL}	V_{pp} setup time		2			ms
t_{AVEL}	Address setup time		2			µs
t_{DVEL}	Data setup time		2			µs
t_{DZOL}	$\overline{OE}$ setup time		2			µs
t_{OHAX}	Address hold time		0			µs
t_{EHDZ}	Data hold time		2			µs
V_{pp}	Programming supply voltage		12.2	12.5	12.8	V
V_{CC}	Programming supply voltage		5.75	6.0	6.25	V
I_{PP2}	V_{CC} supply current (program) $\overline{CE} = V_{IL}$				50	mA
V_{IH}	High level input voltage		2.0		V_{CC}-1	V
V_{IL}	Low level input voltage		-0.1		0.8	V

1/ Initial program pulse width tolerance is 1 ms +/-5%.

2/ The length of the overprogram pulse width may vary from 2.8 ms to 78.75 ms as a function of iteration counter value X.

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TABLE V. Input/output conditions.

For table III subgroups 9, 10, 11, 12.

Symbol	Terminatons	A	B	Unit
V_{CC}	V_{CC}	4.5	5.5	V
V_{IH}	Logic inputs	2.4	2.4	V
V_{IL}	Logic inputs	0.45	0.45	V
V_{OH} 1/	Logic outputs	2.0	2.0	V
V_{OL} 1/	Logic outputs	0.8	0.8	V

1/ Output compare levels for subgroups 9, 10 and 11.

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ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
0	55000000000000000000000000000000
90	00000000000000810000000000000000
B0	00000000000000000000000000000081
D0	00000000008100000000000000000000
F0	00000000000000000000000008100000
180	00000000000000000000000000000081
1A0	00000000000008100000000000000000
1C0	00000000000000000000000008100000
1E0	00000000008100000000000000000000
290	00000000000000000000000000008100
2B0	00000000000081000000000000000000
2D0	00000000000000000000008100000000
2F0	00000000810000000000000000000000
380	00000000000081000000000000000000
3A0	00000000000000000000000000008100
3C0	00000000810000000000000000000000
3E0	00000000000000000000008100000000
410	00000000C0000081000000000000000000
430	00000000000000000000000000000081
450	00000000008100000000000000000000
470	00000000000000000000000008100000
500	00000000000000000000000000000081
520	00000000000000810000000000000000
540	00000000000000000000000008100000
560	00000000008100000000000000000000
610	00000000000000000000000000008100
630	00000000000081000000000000000000
650	00000000000000000000008100000000
670	00000000810000000000000000000000
700	00000000000081000000000000000000
720	00000000000000000000000000008100
740	00000000810000000000000000000000
760	00000000000000000000008100000000
890	00000081000000000000000000000000
8B0	00000000000000000000008100000000
8D0	00810000000000000000000000000000
8F0	00000000000000000008100000000000
980	00000000000000000000008100000000
9A0	00000081000000000000000000000000
9C0	00000000000000000081000000000000
9E0	00810000000000000000000000000000
A80	00000000000000000008100000000000
AB0	00008100000000000000000000000000
AD0	00000000000000008100000000000000
AF0	81000000000000000000000000000000
B80	00008100000000000000000000000000
BA0	00000000000000000008100000000000
BC0	81000000000000000000000000000000
BE0	00000000000000008100000000000000
C10	00000081000000000000000000000000
C30	00000000000000000000008100000000
C50	00810000000000000000000000000000
C70	00000000000000000081000000000000
D00	00000000000000000008100000000000
D20	00000081000000000000000000000000
D40	00000000000000000081000000000000
D60	00810000000000000000000000000000
E10	00000000000000000008100000000000
E30	00008100000000000000000000000000

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
E50	00000000000000008100000000000000
E70	81000000000000000000000000000000
F00	00008100000000000000000000000000
F20	00000000000000000008100000000000
F40	81000000000000000000000000000000
F60	00000000000000008100000000000000
1090	00000000000000420000000000000000
10B0	00000000000000000000000000000042
10D0	00000000004200000000000000000000
10F0	00000000000000000000000000420000
1180	00000000000000000000000000000042
11A0	00000000000004200000000000000000
11C0	00000000000000000000000000420000
11E0	00000000004200000000000000000000
1290	00000000000000000000000000004200
12B0	00000000000042000000000000000000
12D0	00000000000000000000000042000000
12F0	00000000420000000000000000000000
1380	00000000000042000000000000000000
13A0	00000000000000000000000000004200
13C0	00000000420000000000000000000000
13E0	00000000000000000000000042000000
1410	00000000000000042000000000000000
1430	00000000000000000000000000000042
1450	00000000004200000000000000000000
1470	00000000000000000000000000420000
1500	00000000000000000000000000000042
1520	00000000000000420000000000000000
1540	00000000000000000000000000420000
1560	00000000004200000000000000000000
1610	00000000000000000000000000004200
1630	00000000000042000000000000000000
1650	00000000000000000000000042000000
1670	00000000420000000000000000000000
1700	00000000000042000000000000000000
1720	00000000000000000000000000004200
1740	00000000420000000000000000000000
1760	00000000000000000000000042000000
1890	00000042000000000000000000000000
18B0	00000000000000000000004200000000
18D0	00420000000000000000000000000000
18F0	00000000000000000042000000000000
1980	00000000000000000000004200000000
19A0	00000042000000000000000000000000
19C0	00000000000000000042000000000000
19E0	00420000000000000000000000000000
1A90	00000000000000000004200000000000
1AB0	00004200000000000000000000000000
1AD0	00000000000000004200000000000000
1AF0	42000000000000000000000000000000
1B80	00004200000000000000000000000000
1BA0	00000000000000000042000000000000
1BC0	42000000000000000000000000000000
1BE0	00000000000000004200000000000000
1C10	00000042000000000000000000000000
1C30	00000000000000000000004200000000
1C50	00420000000000000000000000000000
1C70	00000000000000000042000000000000
1D00	00000000000000000000004200000000

FIGURE 7. Data pattern.

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ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
1D20	0000004200000000000000000000000000
1D40	0000000000000000004200000000000000
1D60	0042000000000000000000000000000000
1E10	0000000000000000000042000000000000
1E30	0000420000000000000000000000000000
1E50	0000000000000000420000000000000000
1E70	4200000000000000000000000000000000
1F00	0000420000000000000000000000000000
1F20	0000000000000000000420000000000000
1F40	4200000000000000000000000000000000
1F60	0000000000000000420000000000000000
2090	0000000000000024000000000000000000
20B0	0000000000000000000000000000000024
20D0	0000000000240000000000000000000000
20F0	0000000000000000000000000240000000
2180	0000000000000000000000000000000024
21A0	0000000000000024000000000000000000
21C0	0000000000000000000000000240000000
21E0	0000000000240000000000000000000000
2290	0000000000000000000000000000002400
22B0	0000000000002400000000000000000000
22D0	0000000000000000000000024000000000
22F0	0000000024000000000000000000000000
2380	0000000000002400000000000000000000
23A0	0000000000000000000000000002400000
23C0	0000000024000000000000000000000000
23E0	0000000000000000000000024000000000
2410	0000000000000024000000000000000000
2430	0000000000000000000000000000000024
2450	0000000000240000000000000000000000
2470	0000000000000000000000000240000000
2500	0000000000000000000000000000000024
2520	0000000000000024000000000000000000
2540	0000000000000000000000000240000000
2560	0000000000240000000000000000000000
2610	0000000000000000000000000002400000
2630	0000000000002400000000000000000000
2650	0000000000000000000000024000000000
2670	0000000024000000000000000000000000
2700	0000000000002400000000000000000000
2720	0000000000000000000000000002400000
2740	0000000024000000000000000000000000
2760	0000000000000000000000024000000000
2890	0000002400000000000000000000000000
28B0	0000000000000000000002400000000000
28D0	0024000000000000000000000000000000
28F0	0000000000000000002400000000000000
2980	0000000000000000000000024000000000
29A0	0000002400000000000000000000000000
29C0	0000000000000000024000000000000000
29E0	0024000000000000000000000000000000
2A90	0000000000000000000240000000000000
2AB0	0000240000000000000000000000000000
2AD0	0000000000000000240000000000000000
2AF0	2400000000000000000000000000000000
2B80	0000240000000000000000000000000000
2BA0	0000000000000000000240000000000000
2BC0	2400000000000000000000000000000000
2BE0	0000000000000000240000000000000000

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
2C10	0000002400000000000000000000000000
2C30	0000000000000000000002400000000000
2C50	0024000000000000000000000000000000
2C70	0000000000000000002400000000000000
2D00	0000000000000000000000002400000000
2D20	0000002400000000000000000000000000
2D40	0000000000000000002400000000000000
2D60	0024000000000000000000000000000000
2E10	0000000000000000002400000000000000
2E30	0000240000000000000000000000000000
2E50	0000000000000000240000000000000000
2E70	2400000000000000000000000000000000
2F00	0000240000000000000000000000000000
2F20	0000000000000000002400000000000000
2F40	2400000000000000000000000000000000
2F80	0000000000000000240000000000000000
3090	0000000000000018000000000000000000
30B0	0000000000000000000000000000000018
30D0	0000000000180000000000000000000000
30F0	0000000000000000000000000000180000
3180	0000000000000000000000000000000018
31A0	0000000000000018000000000000000000
31C0	0000000000000000000000000000180000
31E0	0000000000180000000000000000000000
3290	0000000000000000000000000000001800
32B0	0000000000001800000000000000000000
32D0	0000000000000000000000000018000000
32F0	0000000018000000000000000000000000
3380	0000000000001800000000000000000000
33A0	0000000000000000000000000000001800
33C0	0000000018000000000000000000000000
33E0	0000000000000000000000000018000000
3410	0000000000000018000000000000000000
3430	0000000000000000000000000000000018
3450	0000000000180000000000000000000000
3470	0000000000000000000000000000180000
3500	0000000000000000000000000000000018
3520	0000000000000018000000000000000000
3540	0000000000000000000000000000180000
3560	0000000000180000000000000000000000
3610	0000000000000000000000000000001800
3630	0000000000001800000000000000000000
3650	0000000000000000000000000018000000
3670	0000000018000000000000000000000000
3700	0000000000001800000000000000000000
3720	0000000000000000000000000000001800
3740	0000000018000000000000000000000000
3760	0000000000000000000000000018000000
3890	0000001800000000000000000000000000
38B0	0000000000000000000000001800000000
38D0	0018000000000000000000000000000000
38F0	0000000000000000001800000000000000
3980	0000000000000000000000001800000000
39A0	0000001800000000000000000000000000
39C0	0000000000000000001800000000000000
39E0	0018000000000000000000000000000000
3A90	0000000000000000000001800000000000
3AB0	0000180000000000000000000000000000
3AD0	0000000000000000180000000000000000

FIGURE 7. Data pattern - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:54Z
Check the source to verify that this is the current version before use.

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
4000	00000000000000001800000000000000
4020	18000000000000000000000000000000
4040	00000000000000000000180000000000
4060	00001800000000000000000000000000
4110	18000000000000000000000000000000
4130	00000000000000001800000000000000
4150	00001800000000000000000000000000
4170	00000000000000000000180000000000
4200	00180000000000000000000000000000
4220	00000000000000000018000000000000
4240	00000018000000000000000000000000
4260	00000000000000000000001800000000
4310	00000000000000000018000000000000
4330	00180000000000000000000000000000
4350	00000000000000000000001800000000
4370	00000018000000000000000000000000
4480	00000000000000001800000000000000
44A0	18000000000000000000000000000000
44C0	00000000000000000000180000000000
44E0	00001800000000000000000000000000
4590	18000000000000000000000000000000
45B0	00000000000000001800000000000000
45D0	00001800000000000000000000000000
45F0	00000000000000000018000000000000
4680	00180000000000000000000000000000
46A0	00000000000000000018000000000000
46C0	00000018000000000000000000000000
46E0	00000000000000000000001800000000
4790	00000000000000000018000000000000
47B0	00180000000000000000000000000000
47D0	00000000000000000000001800000000
47F0	00000018000000000000000000000000
4800	00000000000000000000000018000000
4820	00000000180000000000000000000000
4840	00000000000000000000000000001800
4860	00000000000018000000000000000000
4910	00000000180000000000000000000000
4930	00000000000000000000000018000000
4950	00000000000018000000000000000000
4970	00000000000000000000000000001800
4A00	00000000001800000000000000000000
4A20	00000000000000000000000000180000
4A40	00000000000000180000000000000000
4A60	00000000000000000000000000000018
4B10	00000000000000000000000000180000
4B30	00000000001800000000000000000000
4B50	00000000000000000000000000000018
4B70	00000000000000180000000000000000
4C80	00000000000000000000000018000000
4CA0	00000000180000000000000000000000
4CC0	00000000000000000000000000001800
4CE0	00000000000018000000000000000000
4D90	00000000180000000000000000000000
4DB0	00000000000000000000000018000000
4DD0	00000000000018000000000000000000
4DF0	00000000000000000000000000001800
4E80	00000000001800000000000000000000
4EA0	00000000000000000000000000180000
4EC0	00000000000000180000000000000000

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
3AF0	18000000000000000000000000000000
3B80	00001800000000000000000000000000
3BA0	00000000000000000000180000000000
3BC0	18000000000000000000000000000000
3BE0	00000000000000001800000000000000
3C10	00000018000000000000000000000000
3C30	00000000000000000000001800000000
3C50	00180000000000000000000000000000
3C70	00000000000000000018000000000000
3D00	00000000000000000000001800000000
3D20	00000018000000000000000000000000
3D40	00000000000000001800000000000000
3D60	00180000000000000000000000000000
3E10	00000000000000000018000000000000
3E30	00001800000000000000000000000000
3E50	00000000000000001800000000000000
3E70	18000000000000000000000000000000
3F00	00001800000000000000000000000000
3F20	00000000000000000018000000000000
3F40	18000000000000000000000000000000
3F60	00000000000000001800000000000000
4EE0	00000000000000000000000000000018
4F90	00000000000000000000000000180000
4FB0	00000000001800000000000000000000
4FD0	00000000000000000000000000000018
4FF0	00000000000000180000000000000000
5000	00000000000000002400000000000000
5020	24000000000000000000000000000000
5040	00000000000000000000240000000000
5060	00002400000000000000000000000000
5110	24000000000000000000000000000000
5130	00000000000000002400000000000000
5150	00002400000000000000000000000000
5170	00000000000000000000240000000000
5200	00240000000000000000000000000000
5220	00000000000000000024000000000000
5240	00000024000000000000000000000000
5260	00000000000000000000002400000000
5310	00000000000000000024000000000000
5330	00240000000000000000000000000000
5350	00000000000000000000002400000000
5370	00000024000000000000000000000000
5480	00000000000000002400000000000000
54A0	24000000000000000000000000000000
54C0	00000000000000000002400000000000
54E0	00002400000000000000000000000000
5590	24000000000000000000000000000000
55B0	00000000000000002400000000000000
55D0	00002400000000000000000000000000
55F0	00000000000000000002400000000000
5680	00240000000000000000000000000000
56A0	00000000000000002400000000000000
56C0	00000024000000000000000000000000
56E0	00000000000000000002400000000000
5790	00000000000000002400000000000000
57B0	00240000000000000000000000000000
57D0	00000000000000000002400000000000
57F0	00000024000000000000000000000000
5800	00000000000000000000002400000000

FIGURE 7. Data pattern - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:54Z
Check the source to verify that this is the current version before use.

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
5820	00000000240000000000000000000000
5840	00000000000000000000000000002400
5860	00000000000024000000000000000000
5910	00000000240000000000000000000000
5930	00000000000000000000000024000000
5950	00000000000024000000000000000000
5970	00000000000000000000000000002400
5A00	00000000002400000000000000000000
5A20	00000000000000000000000000240000
5A40	00000000000002400000000000000000
5A60	00000000000000000000000000000024
5B10	00000000000000000000000000240000
5B30	00000000002400000000000000000000
5B50	00000000000000000000000000000024
5B70	00000000000002400000000000000000
5C80	00000000000000000000000024000000
5CA0	00000000240000000000000000000000
5CC0	00000000000000000000000000002400
5CE0	00000000000024000000000000000000
5D90	00000000240000000000000000000000
5DB0	00000000000000000000000024000000
5DD0	00000000000024000000000000000000
5DF0	00000000000000000000000000002400
5E80	00000000002400000000000000000000
5EA0	00000000000000000000000000240000
5EC0	00000000000002400000000000000000
5EE0	00000000000000000000000000000024
5F90	00000000000000000000000000240000
5FB0	00000000002400000000000000000000
5FD0	00000000000000000000000000000024
5FF0	00000000000002400000000000000000
6000	00000000000000004200000000000000
6020	42000000000000000000000000000000
6040	00000000000000000000420000000000
6060	00004200000000000000000000000000
6110	42000000000000000000000000000000
6130	00000000000000004200000000000000
6150	00004200000000000000000000000000
6170	00000000000000000000420000000000
6200	00420000000000000000000000000000
6220	00000000000000000042000000000000
6240	00000042000000000000000000000000
6260	00000000000000000000004200000000
6310	00000000000000000042000000000000
6330	00420000000000000000000000000000
6350	00000000000000000000004200000000
6370	00000042000000000000000000000000
6480	00000000000000004200000000000000
64A0	42000000000000000000000000000000
64C0	00000000000000000000420000000000
64E0	00004200000000000000000000000000
6590	42000000000000000000000000000000
65B0	00000000000000004200000000000000
65D0	00004200000000000000000000000000
65F0	00000000000000000000420000000000
6680	00420000000000000000000000000000
66A0	00000000000000000042000000000000
66C0	00000042000000000000000000000000
66E0	00000000000000000000004200000000

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
6790	00000000000000000042000000000000
67B0	00420000000000000000000000000000
67D0	00000000000000000000004200000000
67F0	00000042000000000000000000000000
6800	00000000000000000000000042000000
6820	00000000420000000000000000000000
6840	00000000000000000000000000004200
6860	00000000000042000000000000000000
6910	00000000420000000000000000000000
6930	00000000000000000000000042000000
6950	00000000000042000000000000000000
6970	00000000000000000000000000004200
6A00	00000000004200000000000000000000
6A20	00000000000000000000000000420000
6A40	00000000000000042000000000000000
6A60	00000000000000000000000000000042
6B10	00000000000000000000000000420000
6B30	00000000004200000000000000000000
6B50	00000000000000000000000000000042
6B70	00000000000004200000000000000000
6C80	00000000000000000000000042000000
6CA0	00000000420000000000000000000000
6CC0	00000000000000000000000000004200
6CE0	00000000000042000000000000000000
6D90	00000000420000000000000000000000
6DB0	00000000000000000000000042000000
6DD0	00000000000042000000000000000000
6DF0	00000000000000000000000000004200
6E80	00000000004200000000000000000000
6EA0	00000000000000000000000000420000
6EC0	00000000000000042000000000000000
6EE0	00000000000000000000000000000042
6F90	00000000000000000000000000420000
6FB0	00000000004200000000000000000000
6FD0	00000000000000000000000000000042
6FF0	00000000000000042000000000000000
7000	00000000000000008100000000000000
7020	81000000000000000000000000000000
7040	00000000000000000008100000000000
7060	00008100000000000000000000000000
7110	81000000000000000000000000000000
7130	00000000000000008100000000000000
7150	00008100000000000000000000000000
7170	00000000000000000008100000000000
7200	00810000000000000000000000000000
7220	00000000000000000810000000000000
7240	00000081000000000000000000000000
7260	00000000000000000008100000000000
7310	00000000000000000810000000000000
7330	00810000000000000000000000000000
7350	00000000000000000000008100000000
7370	00000081000000000000000000000000
73E0	FF00FF00FF00FF0000FF00FF00FF00FF
73F0	00FF00FF00FF00FFFF00FF00FF00FF00
7480	00000000000000008100000000000000
74A0	81000000000000000000000000000000
74C0	00000000000000000008100000000000
74E0	00008100000000000000000000000000
7590	81000000000000000000000000000000

FIGURE 7. Data pattern - Continued.

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ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
75B0	00000000000000008100000000000000
75D0	00008100000000000000000000000000
75F0	00000000000000000000810000000000
7680	00810000000000000000000000000000
76A0	00000000000000000081000000000000
76C0	00000081000000000000000000000000
76E0	00000000000000000000008100000000
7790	00000000000000000081000000000000
77B0	00810000000000000000000000000000
77D0	00000000000000000000008100000000
77E0	FF00FF00FF00FF0000FF00FF00FF00FF
77F0	00FF00FF00FF00FFFF00FF00FF00FF00
7800	00000000000000000000000081000000
7820	00000000810000000000000000000000
7840	00000000000000000000000000008100
7860	00000000000081000000000000000000
7910	00000000810000000000000000000000
7930	00000000000000000000000081000000
7950	00000000000081000000000000000000
7970	00000000000000000000000000008100
7A00	00000000008100000000000000000000
7A20	00000000000000000000000000810000

ADD:	0 1 2 3 4 5 6 7 8 9 A B C D E F
7A40	00000000000000810000000000000000
7A60	00000000000000000000000000000081
7B10	00000000000000000000000000810000
7B30	00000000008100000000000000000000
7B50	00000000000000000000000000000081
7B70	00000000000000810000000000000000
7C80	00000000000000000000000081000000
7CA0	00000000810000000000000000000000
7CC0	00000000000000000000000000008100
7CE0	00000000000081000000000000000000
7D90	00000000810000000000000000000000
7DB0	00000000000000000000000081000000
7DD0	00000000000081000000000000000000
7DF0	00000000000000000000000000008100
7E80	00000000008100000000000000000000
7EA0	00000000000000000000000000810000
7EC0	00000000000008100000000000000000
7EE0	00000000000000000000000000000081
7F90	00000000000000000000000000810000
7FB0	00000000008100000000000000000000
7FD0	00000000000000000000000000000081
7FF0	00000000000008100000000FF00AA

NOTES:
1. The device shall be programmed with all zeros except for the address locations identified in the pattern list.

FIGURE 7. Data pattern - Continued.

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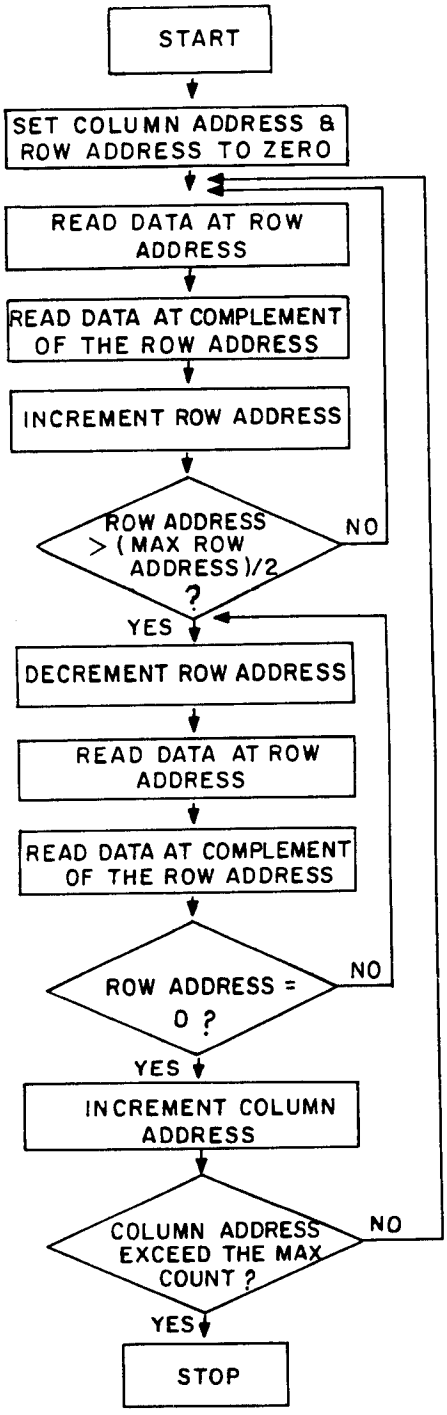


FIGURE 8. Row complement test.

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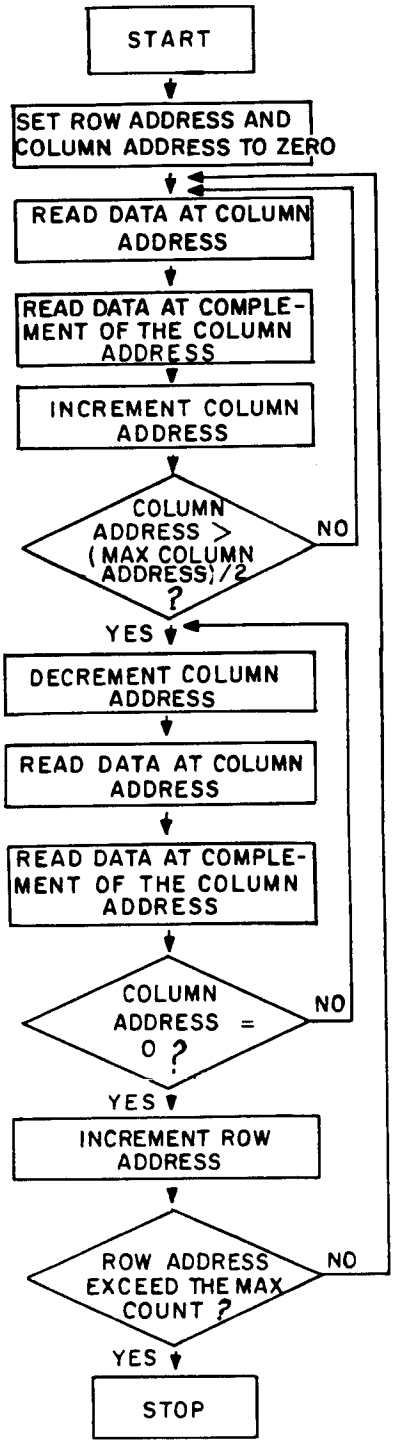


FIGURE 9. Column complement test.

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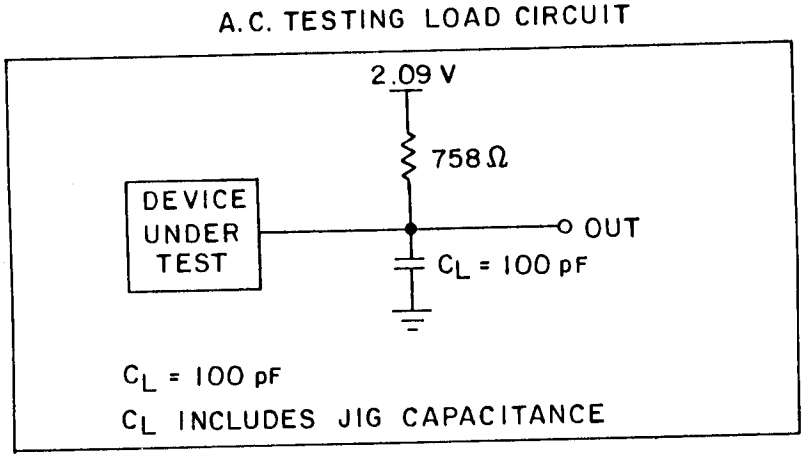


FIGURE 10. Load circuit.

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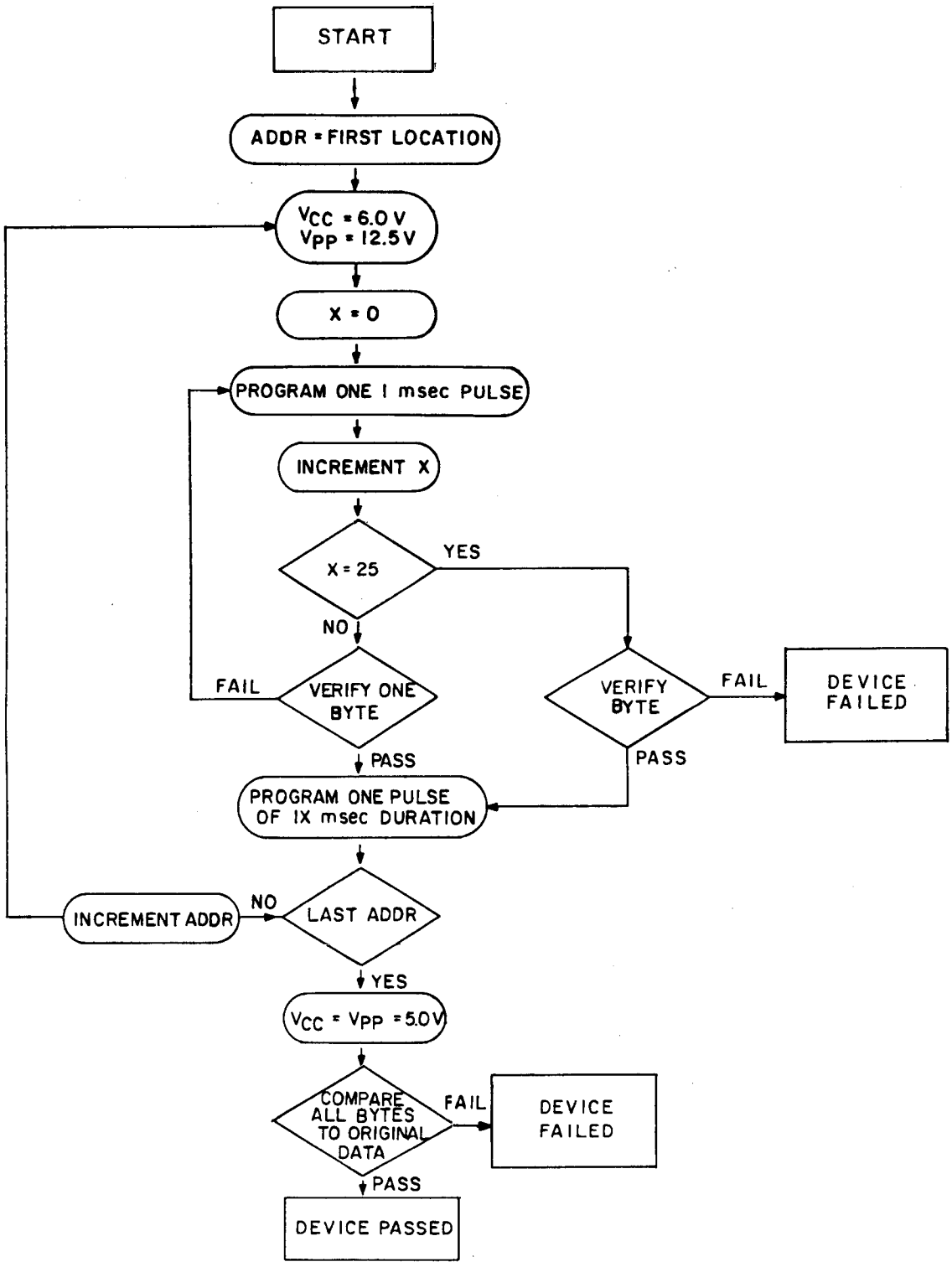


FIGURE 11. Programming flowchart.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{CC}	Supply voltage. Voltage values are with respect to ground, unless otherwise specified, thoughout this detail specification.
V_{PP}	Program supply voltage.
A0 - A14	Address inputs, used to address 1 of 32,768/8 bit locations in static storage array.
$\overline{CE}$	Chip-enable, used along with the output enable (OE) signal to control the state of the data-out terminals.
$\overline{OE}$	Output-enable, used to control the data-out terminals.
O0 - O7	Data input-output, 8 bit wide data bus through which data is stored or accessed.
I_{CC}	Supply current.
I_{PP}	V_{pp} current.
I_{IH}, I_{IL}	Input leakage currents.
I_{OHZ}	High impedance output leakage currents.
V_{IL}	Logical low input voltage.
V_{IH}	Logical high input voltage.
V_{OL}	Logical low output voltage.
V_{OH}	Logical high output voltage.
C_i	Input capacitance.
C_o	Output capacitance.
t_{AVQV}	Address access time.
t_{ELQV}	Chip-enable access time.
t_{OLQV}	Output-enable access time.
t_{AXQX}	Address to data invalid.
t_{AVEL}	Address setup time.
t_{OHAX}	Address hold time.
t_{OHQZ}	Output-enable high to output float.
t_{DVEL}	Data to CE setup time.
t_{EHDZ}	Data hold time.
t_{ELEH}	Program pulse width.
t_{VHEL}	V_{pp} setup time.
t_{EHQZ}	Chip enable high to output float.
t_{PHEL}	V_{CC} setup time.
t_{DZOL}	Output enable setup time.

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6.3.1 Timing parameter abbreviations. All timing abbreviations use lower case characters with upper case character subscripts. The initial character is always t and is followed by four descriptors. These characters specify two signal points arranged in a "from-to" sequence that define a timing interval. The two descriptors for each signal point specify the signal name and the signal transition. Thus the format is:

t X X X X

Signal name from which interval is defined

Transition direction for first signal

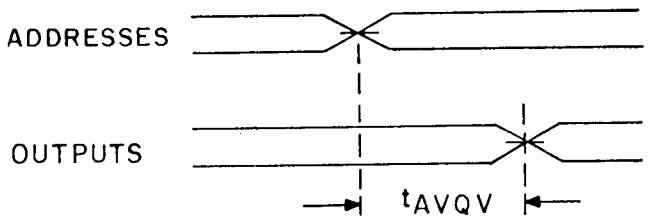
Signal name to which interval is defined

Transition direction for second signal

a. A = Address
D = Data in
Q = Data out
E = Chip-enable
O = Output-enable
P = V_{CC}
V = V_{pp}

b. H = Transition to high
L = Transition to low
V = Transition to valid
X = Transition to invalid or don't care
Z = Transition to off (high impedance)

Example:

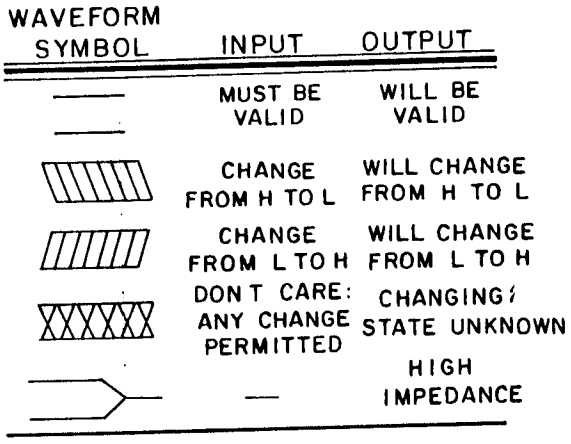


The example shows access time defined at t_{AVQV} time from address valid to output valid.

c. The table of timing values shows either a minimum or a maximum limit for each parameter. Input requirements are specified from the external system point of view. Thus, address set-up time is shown as a minimum since the system must supply at least that much time (even though most devices do not require it). On the other hand, responses from the memory are specified from the device point of view. Thus, the access time is shown as a maximum since the device never provides data later than that time.

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d.



6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Handling. MOS devices shall be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS area.

f. Maintain relative humidity above 50 percent, if practical.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	27256-250
02	27256-200
03	27256-170

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Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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